

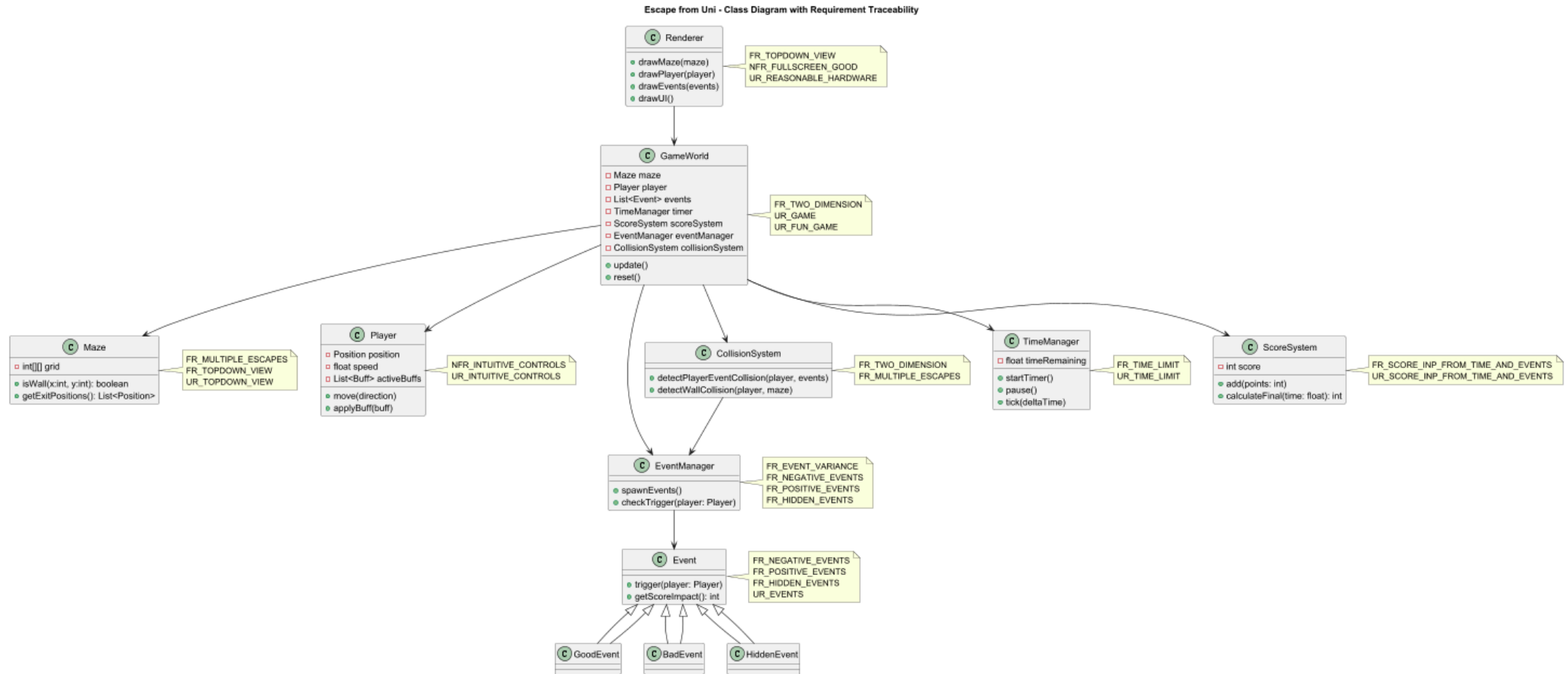
Architecture

CO3 G10

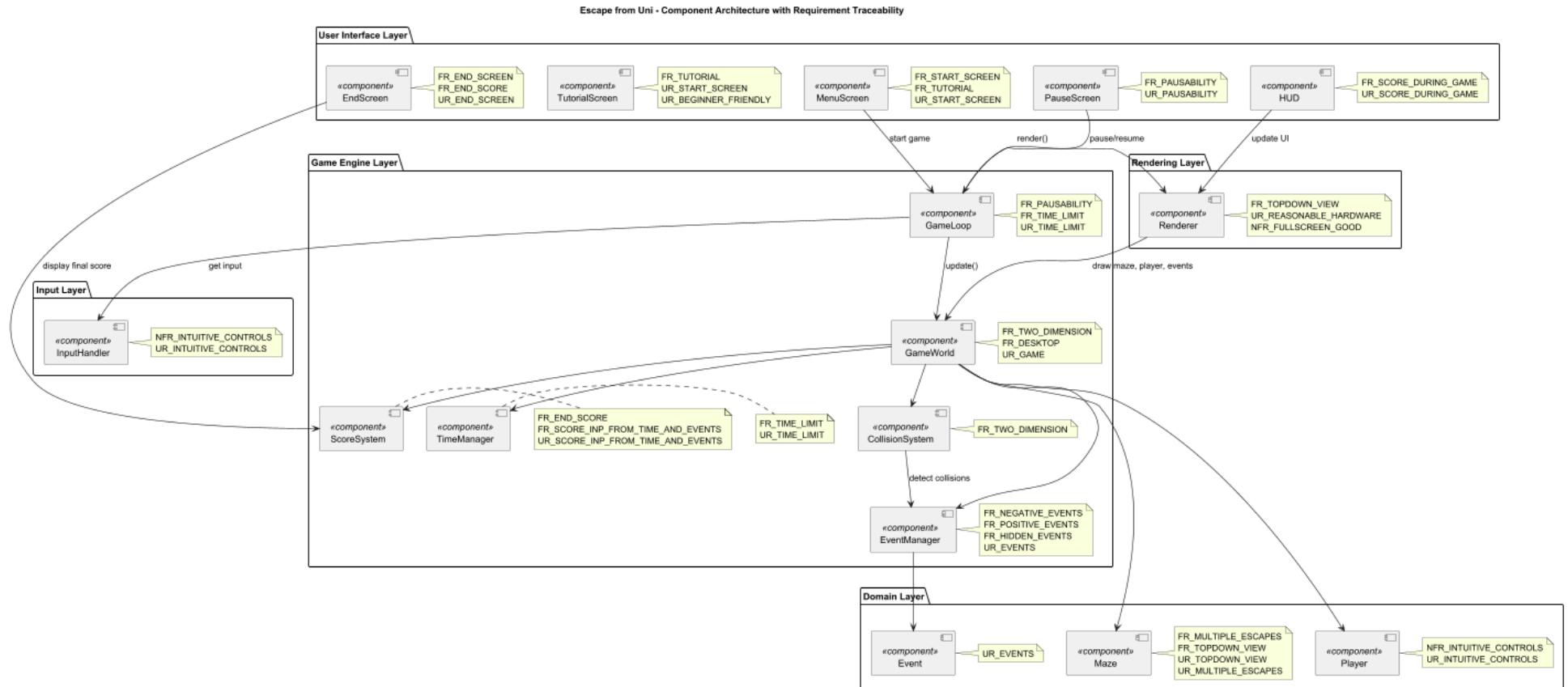
Will Barlow
Harris Byrom
Corbin Hedger
Dmytro Kovalenko
Sharjil Munshi
Oliver Spence
Joshua Thomas
Leon Varghese

All diagrams made with PlantUML.

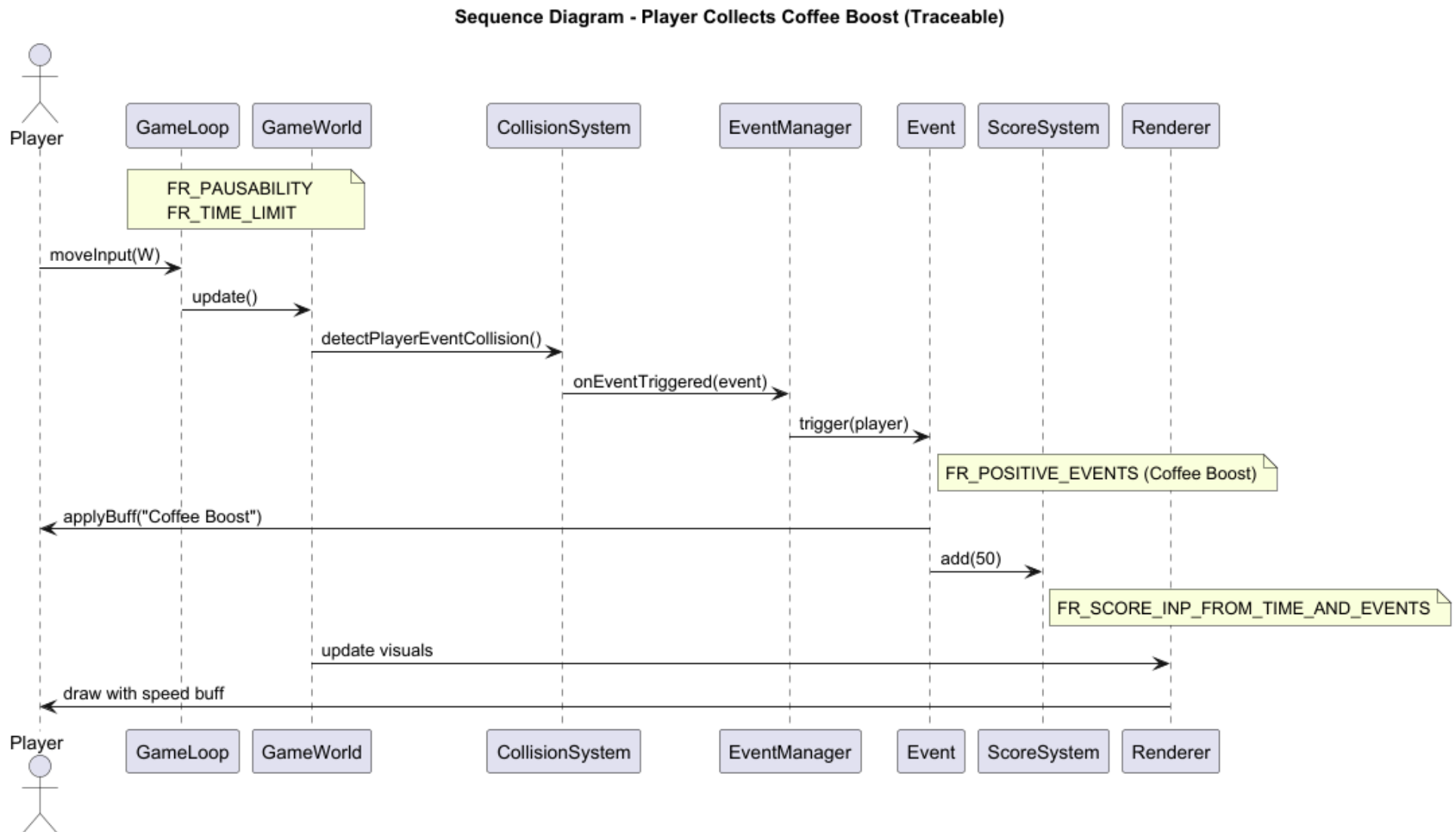
Class Diagram with Requirement Traceability



Component Architecture with Requirement Traceability

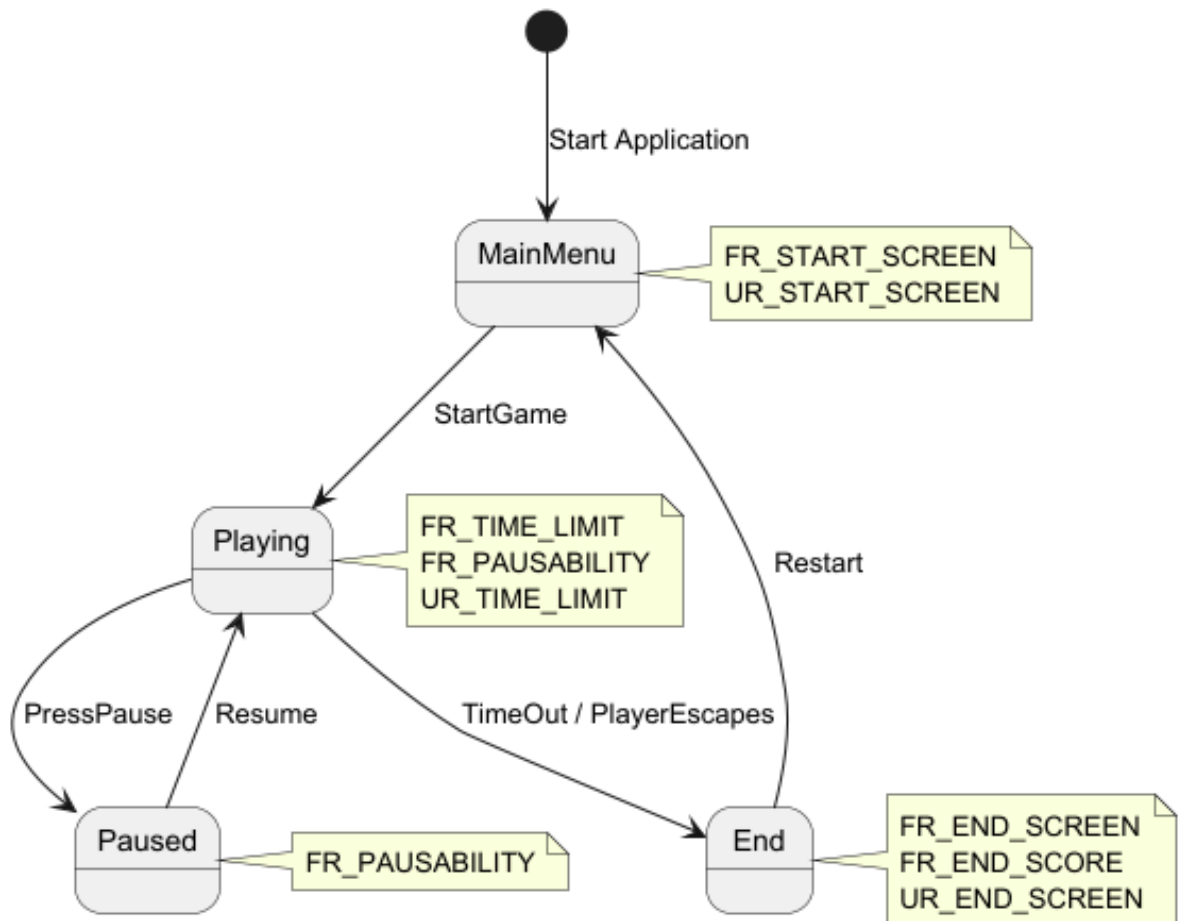


Player Collect Coffee Boost Sequence Diagram



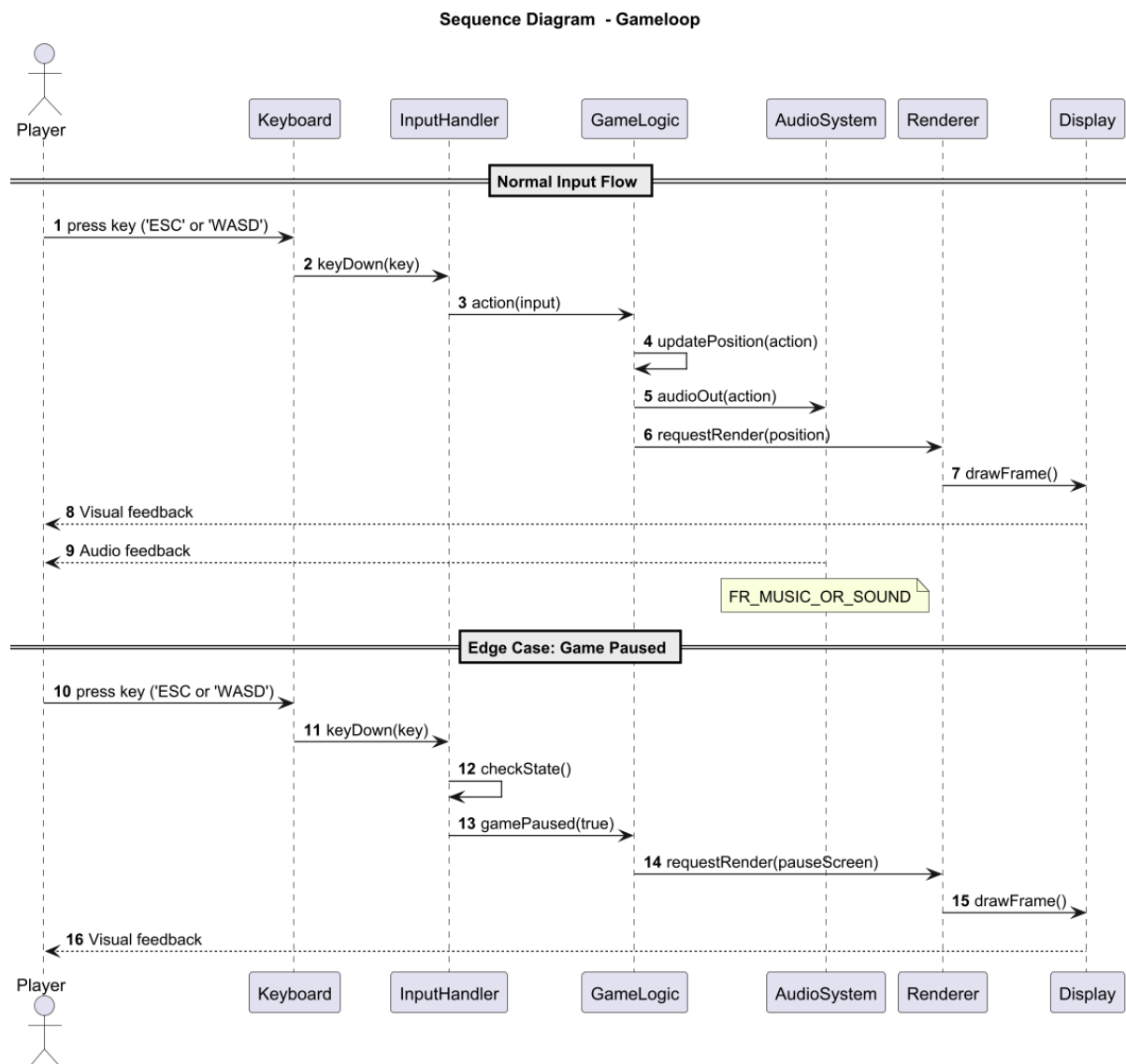
Game State Machine

Game State Machine - Escape from Uni (with Traceability)



Game Loop Diagram

This diagram shows a high level representation of the gameloop that the game follows, from handling inputs to passing them through game logic to apply the effect on the maze entities. We chose to do this so that the processes applied are all encapsulated within the GameLogic (implemented as Game), which helps separate the processing of inputs and outputs so that they can easily be handled, whether it be audio (FR_MUSIC_OR_SOUND) or visual updates. The diagram also shows how GameLogic handles the game being paused (FR_PAUSABILITY), as it uses GameStateHandler to check and change the state, and if the game is paused, keyboard inputs are not used until the user unpauses the game.



Initial design and evolution of architecture